LARYNX

LEARNING OBJECTIVES

At the end of the lecture , the student should be able to know about :

Structure of larynx

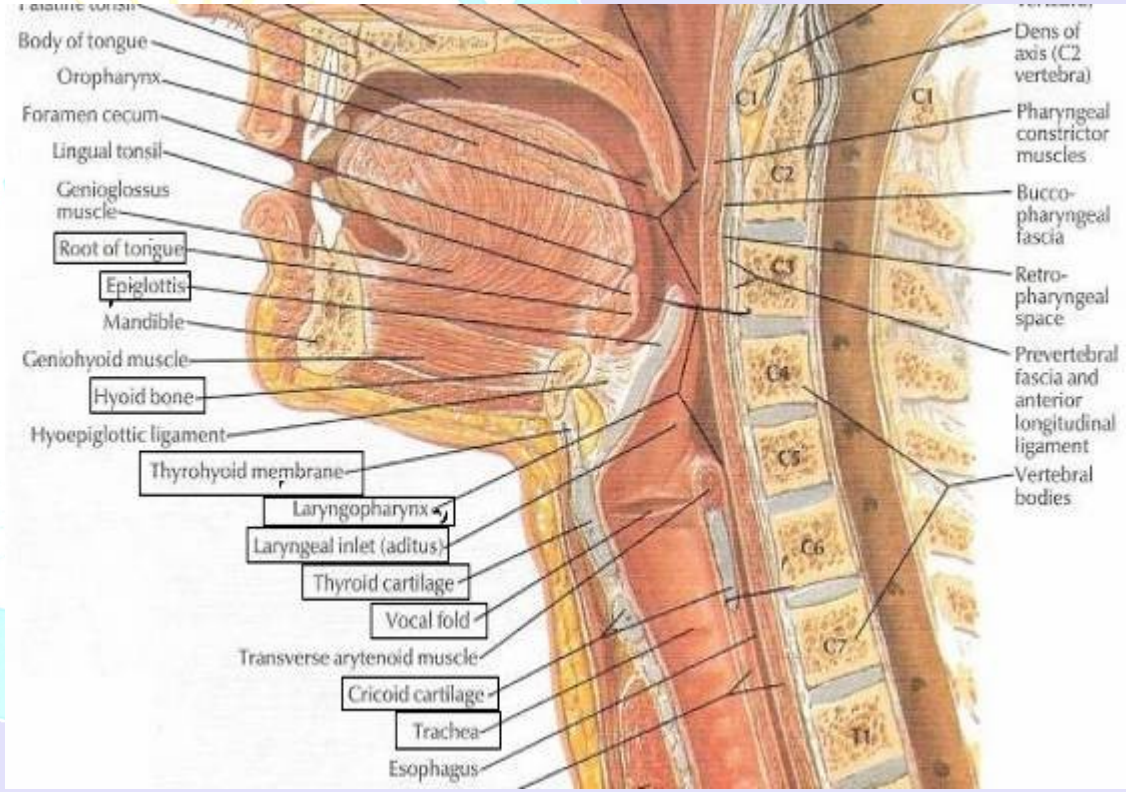
Laryngeal cartilages

Muscles moving vocal cords

Nerve supply of laryngeal muscle

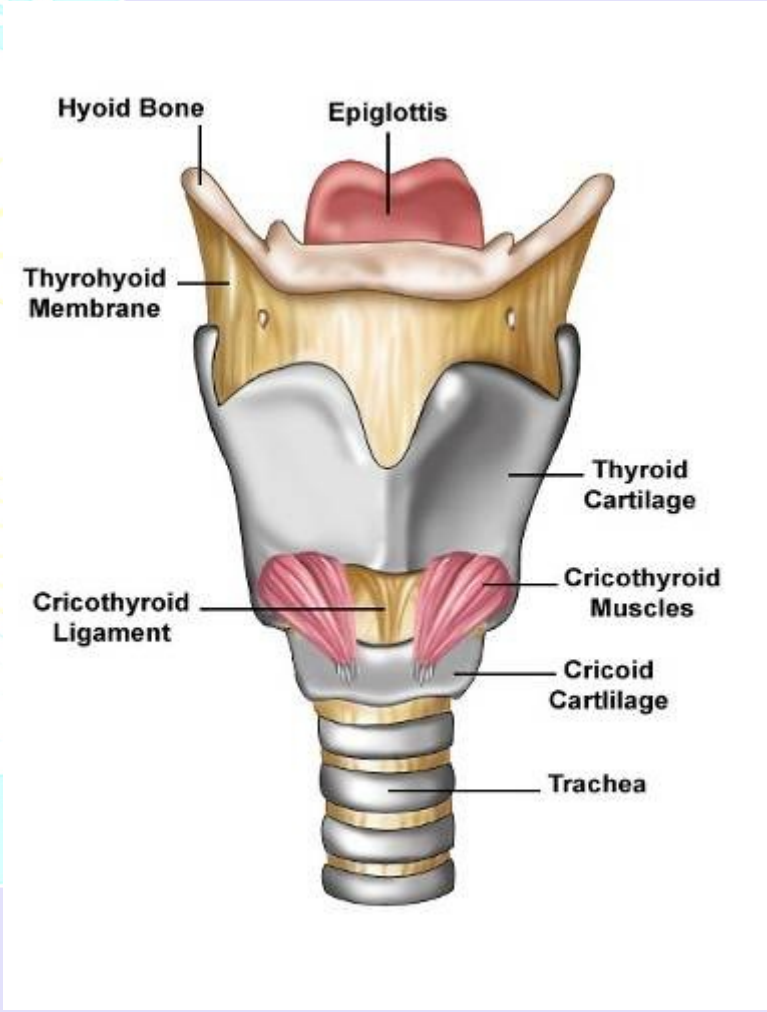
LARYNX

- Musculo ligamentous structure
- Air passage
- Sphincteric device
- Organ of phonation
- Lies in anterior midline of neck from root of tongue to trachea



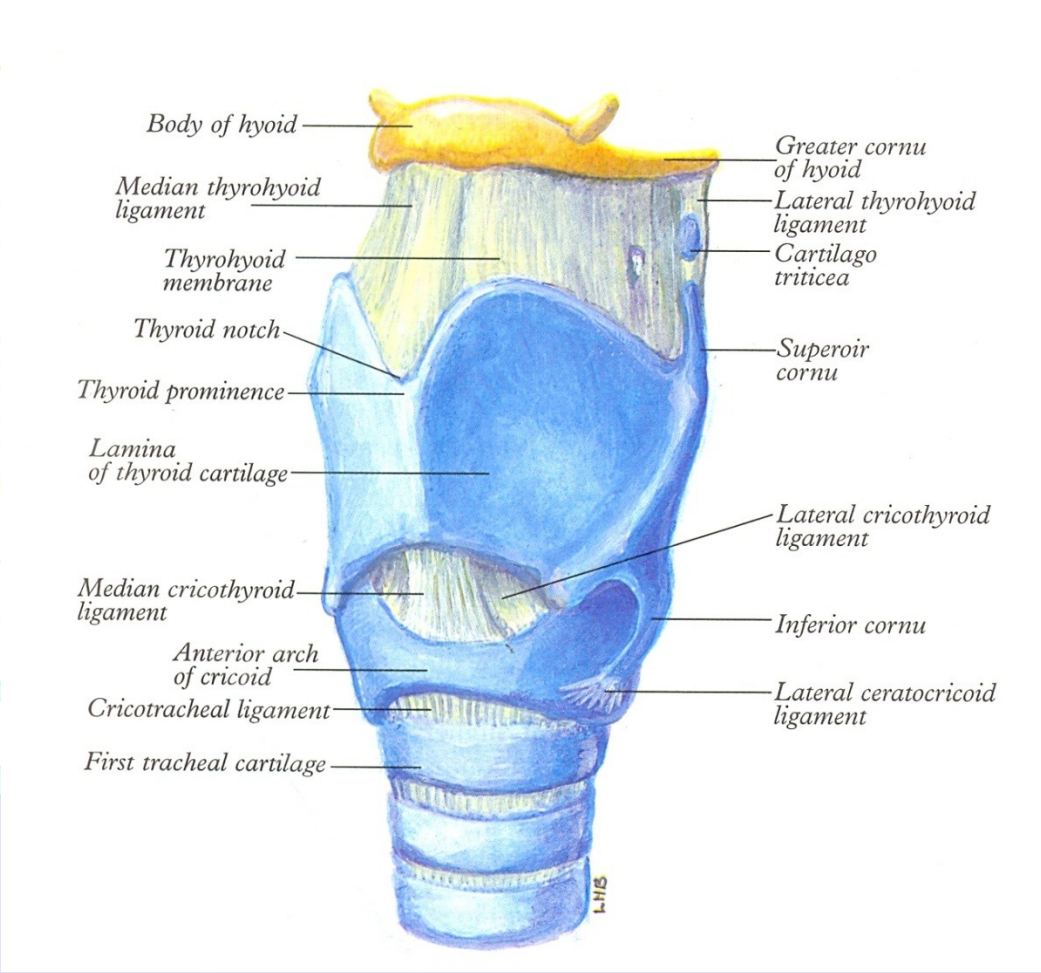
LARYNX

- Open above in laryngopharynx
- Continuous below in trachea
- Lies at the level of C 4, C5, C6 vertebrae
- Covered in front by skin, fascia and infrahyoid muscles
- Covered on sides by thyroid gland
- **Length** 44mm(m), 36mm(f),
- **TS** – 43mm(m), 41mm(f),
- **Sagital** diameter 36m and 26F
- Male larynx enlarges after puberty



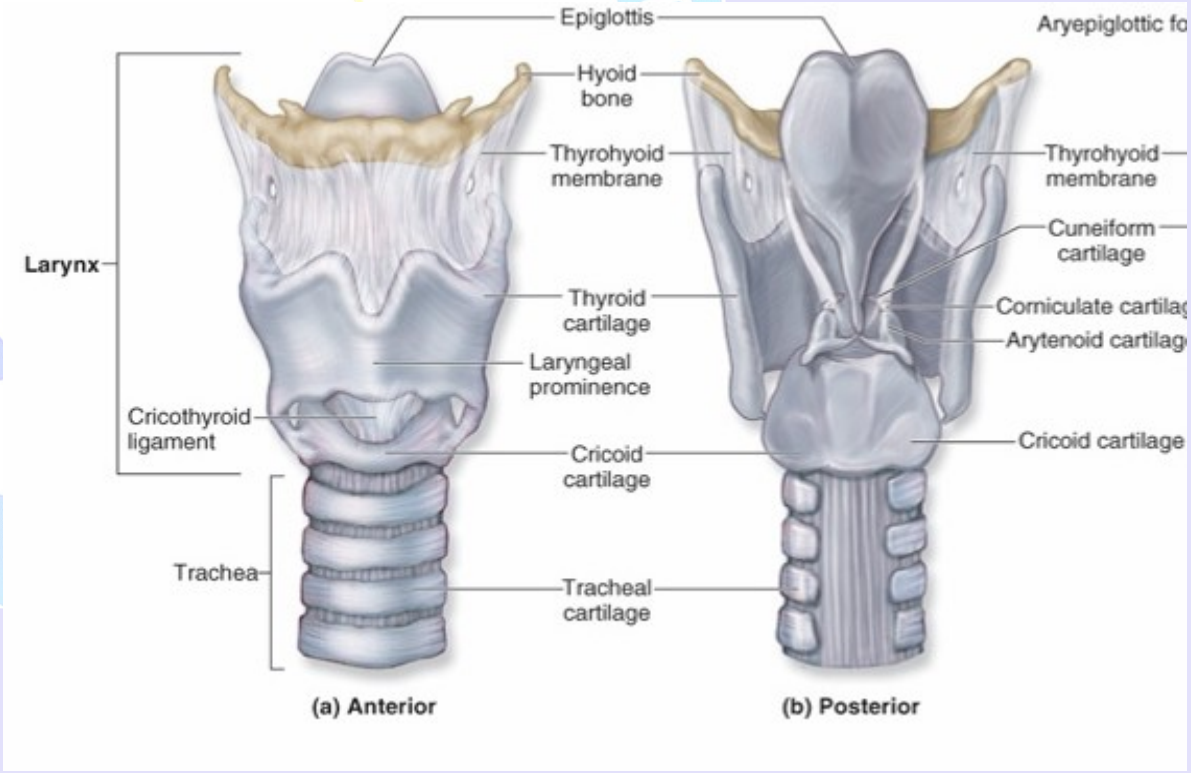
STRUCTURE OF LARYNX

- Cartilagenous framework
  - Paired
  - Unpaired
- ligaments and membranes
  - Extrinsic
  - Intrinsic
- Muscles
  - Extrinsic
  - Intrinsic
- Mucosa
  - Inlet of larynx
  - Cavity of larynx



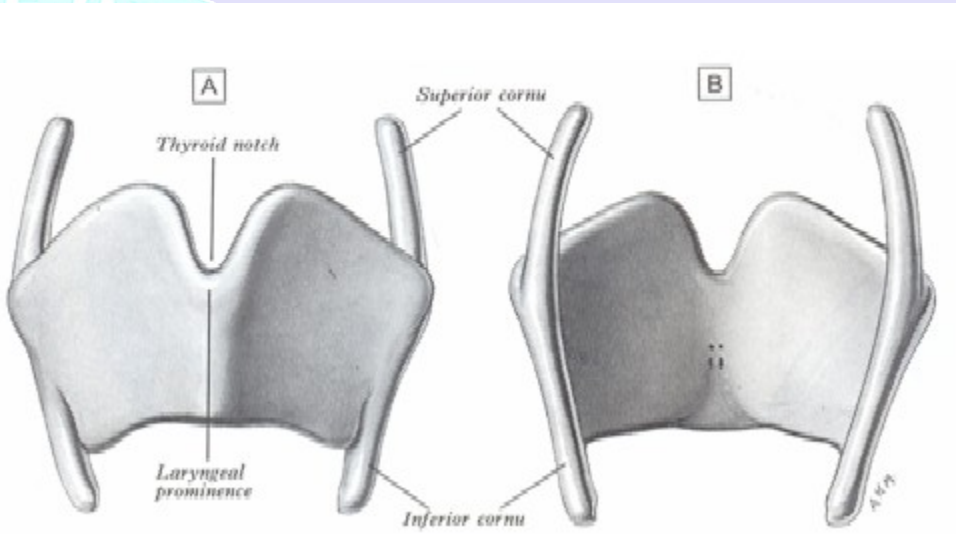
CARTILAGES OF LARYNX

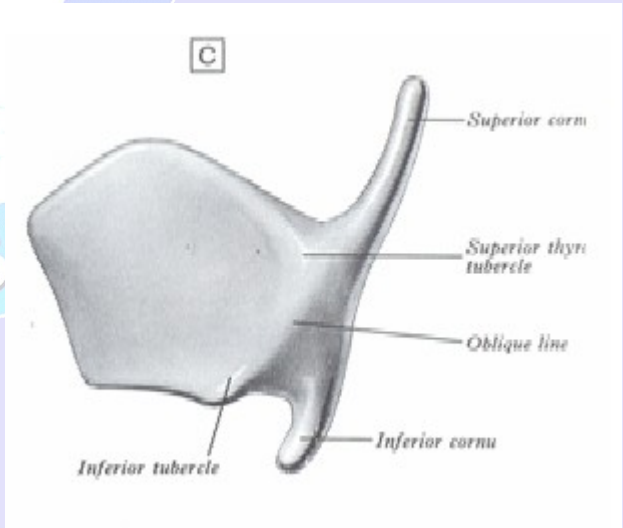
- Unpaired cartilages:
  - Thyroid
  - Cricoid
  - Epiglottis
- Paired cartilages
  - Arytenoid
  - Corniculate
  - Cuneiform



THYROID CARTILAGE

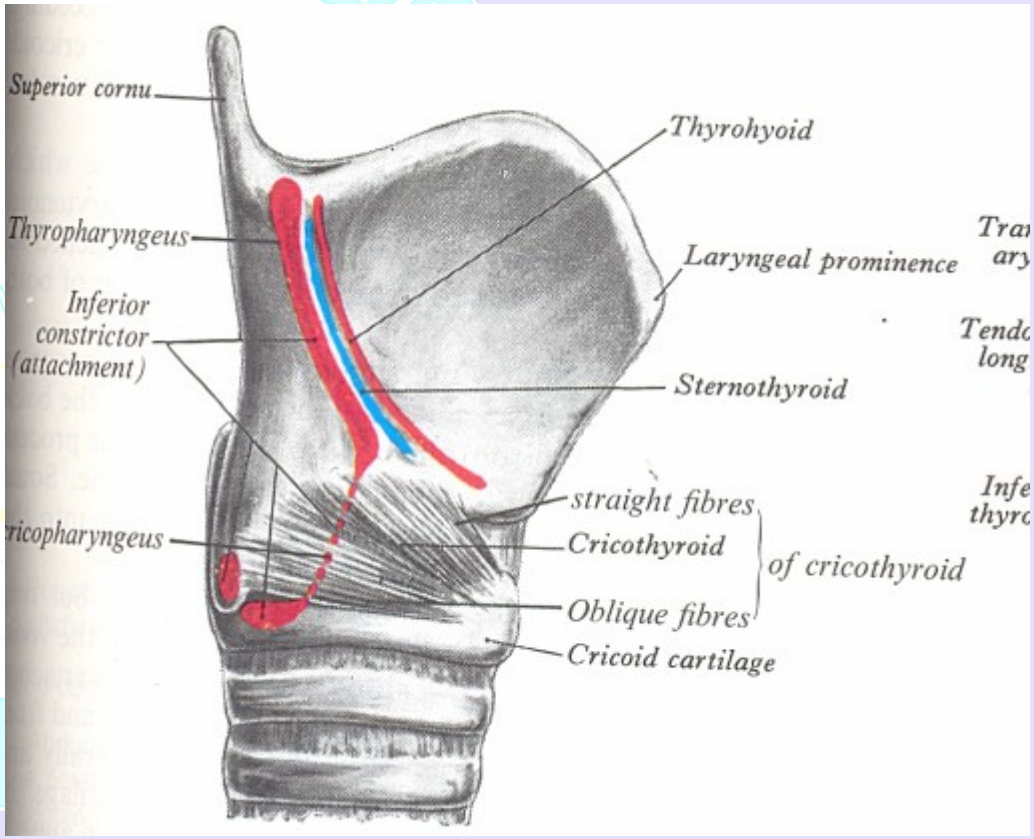
- Largest laryngeal cartilage
- Hyaline
- Two lamina
- Join at midline at V angle
- Thyroid notch between anterior borders of lamina
- Posterior borders extend upward into superior cornu and downwards into inferior cornu
- Anterior surface possess an oblique line





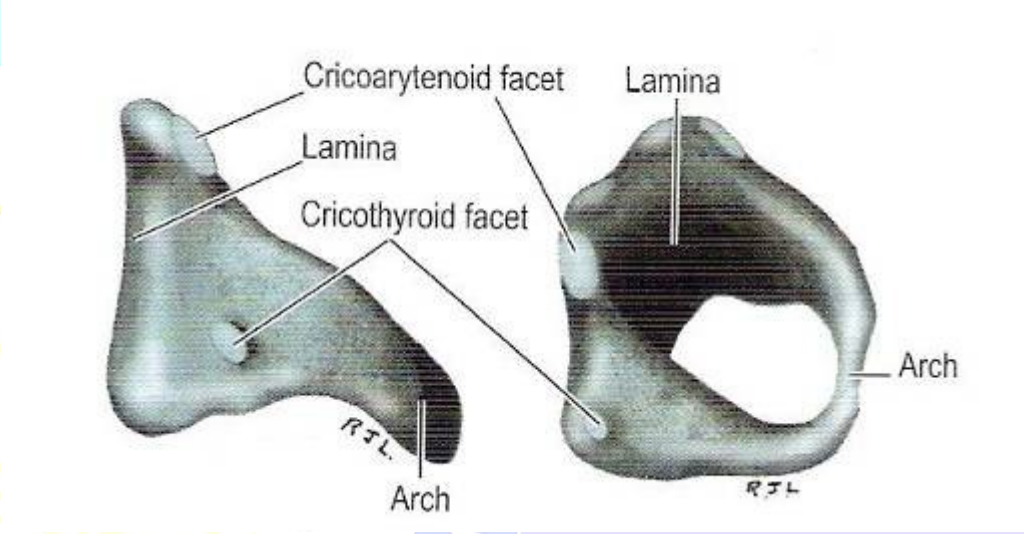
THYROID CARTILAGE
Attachments

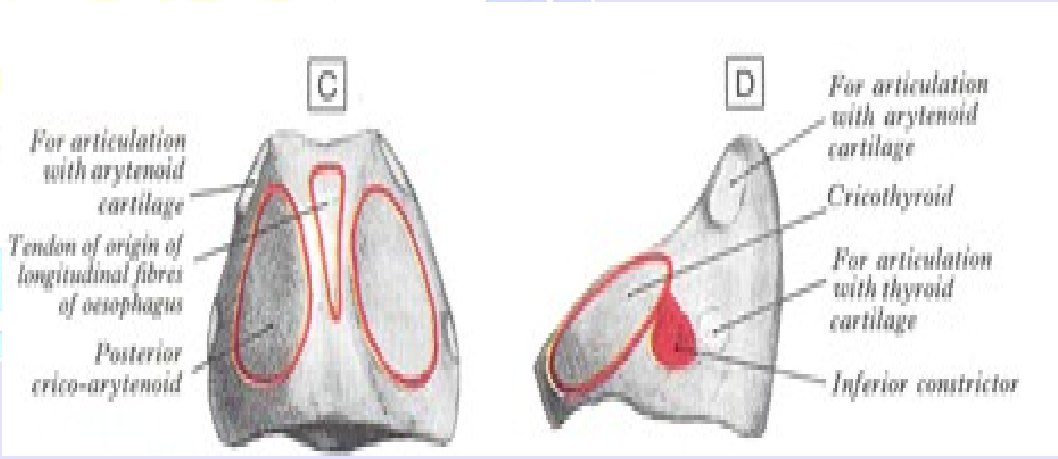
- Superior cornu attacthed with greater horn of hyoid bone
- Inferior cornu articulates with cricoid at cricothyroid joint
- Oblique line provides attachment to
  - inferior constrictor muscle of pharynx
  - Thyrohyoid muscle



CRICOID CARTILAGE

- Ring shaped
- Hyaline cartilage
- Lies below thyroid cartilage
- Broad plate behind, shallow arch in front
- On lateral surface facet for articulation with thyroid
- Posteriorly, on upper border facet for arytenoid

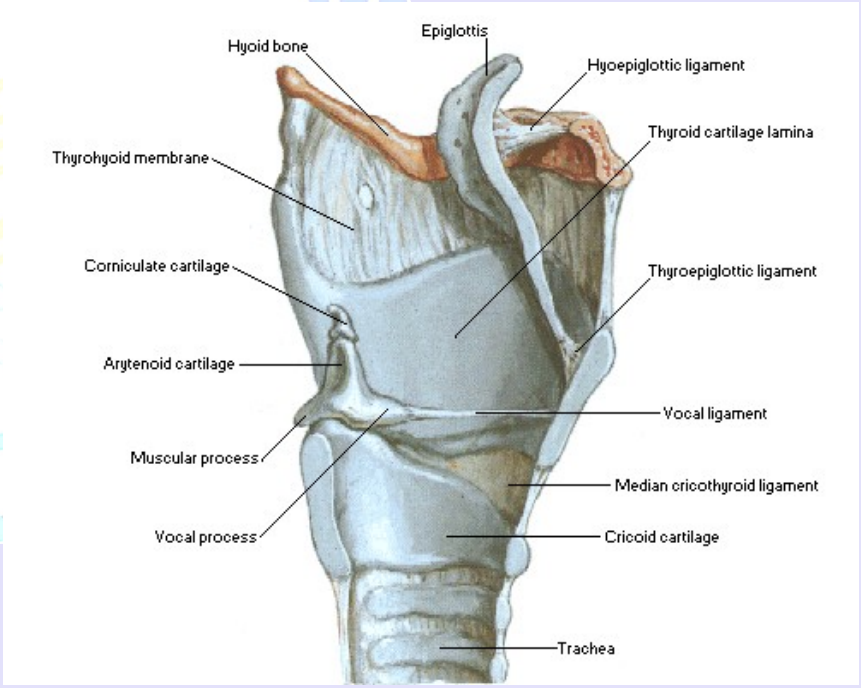




EPIGLOTTIS

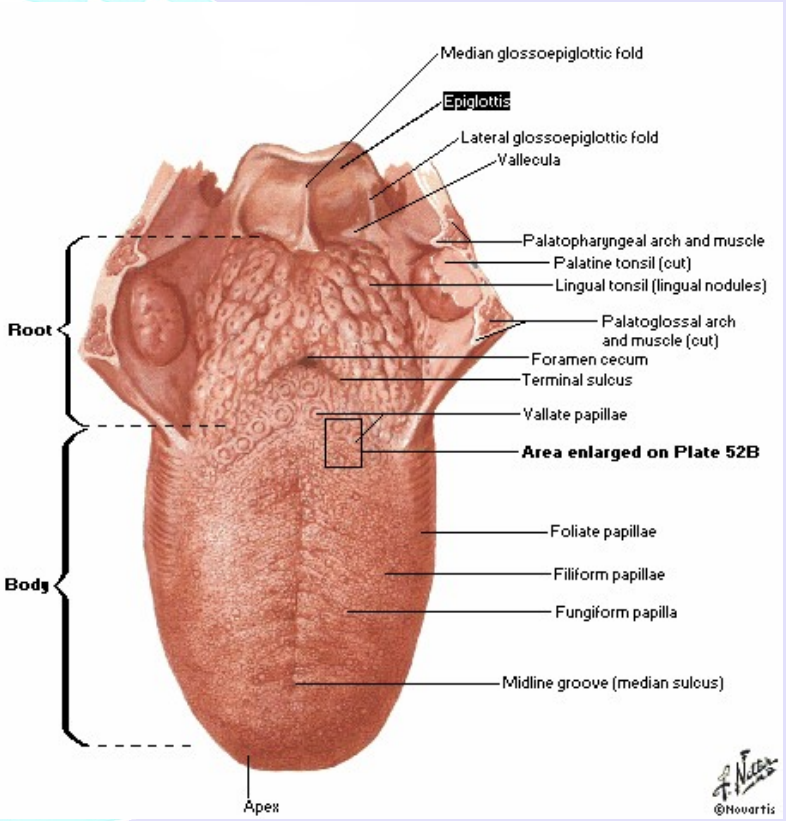
- Leaf shaped
- Elastic cartilage
- Lies behind the root of tongue; in anterior wall of upper pharynx
- Upper end broad and free
- Lower end (stalk) pointed; attached on the posterior surface of angle of thyroid cartilage
- Has anterior and posterior surface





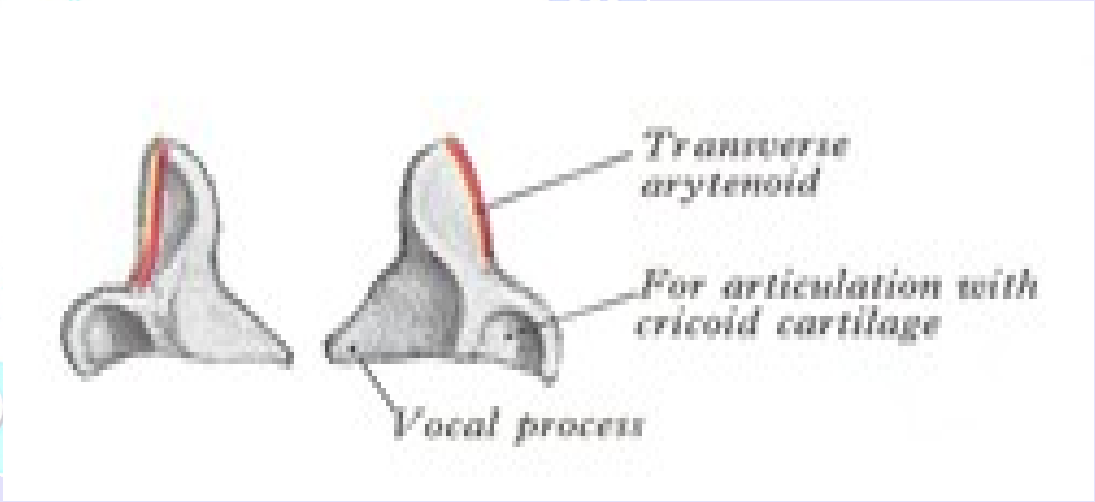
EPIGLOTTIS

- Attached to arytenoid cartilages by aryepiglottic fold
- Covered on both surface by mucosa
- Mucosa passes forward onto tongue as median glossoepiglottic fold
- On each sideis vallecula
- Laterally is lateral glossoepiglottic fold

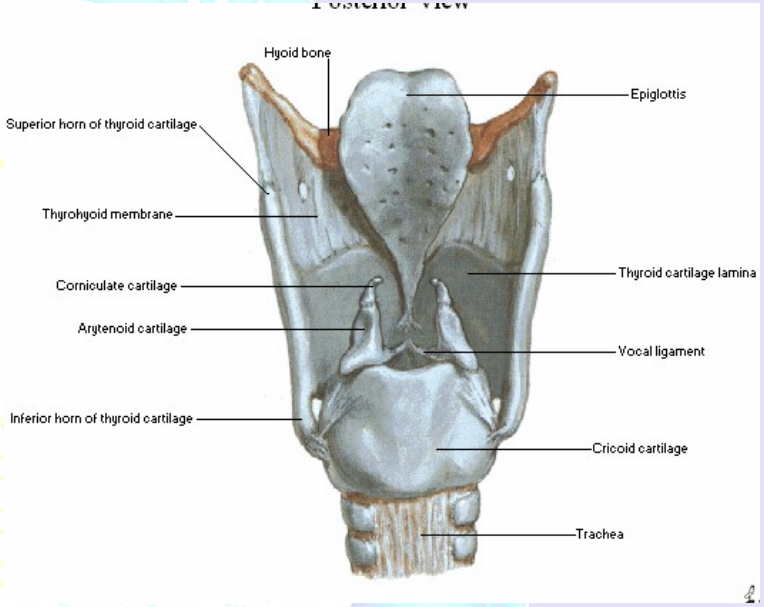


ARYTENOID CARTILAGES

- Paired
- Pyramid shaped
- At the back of larynx
- Only tip is elastic; rest of the body is hyaline
- Possesses :
  - Apex
  - Base
  - Vocal process
  - Muscular process
  - Anterior
  - Medial
  - Posterior surface



- Apex articulates with small corniculate cartilage
- Base articulates with upper border of posterior lamina of cricoid
- Vocal process give attachment to vocal ligament
- Muscular process give attachment to posterior and lateral cricoarytenoid muscle

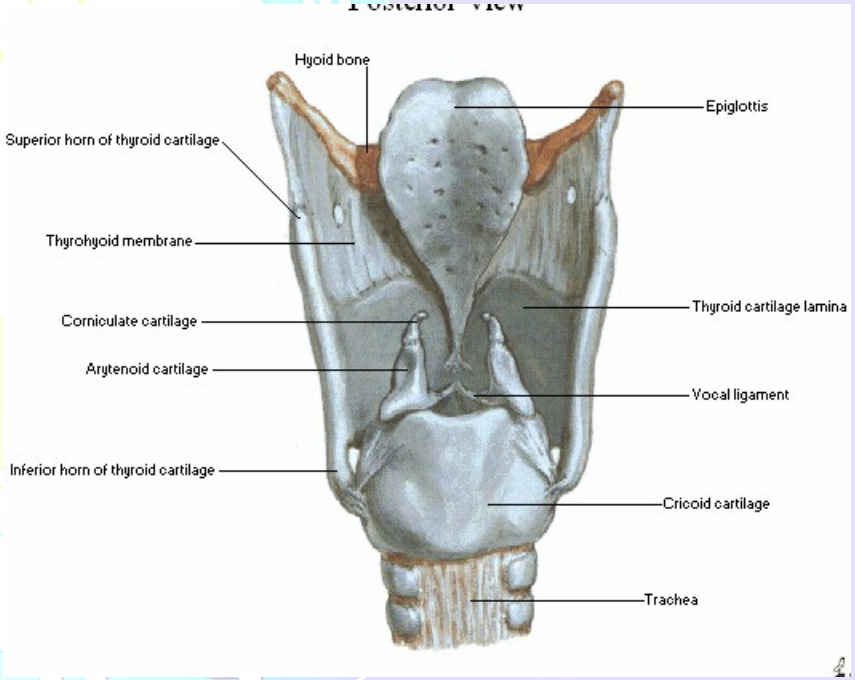


Corniculate:

- Conical nodules
- articulate with the apex of artenoid cartilage
- Attachment to aryepiglottic folds

Cuneiform:

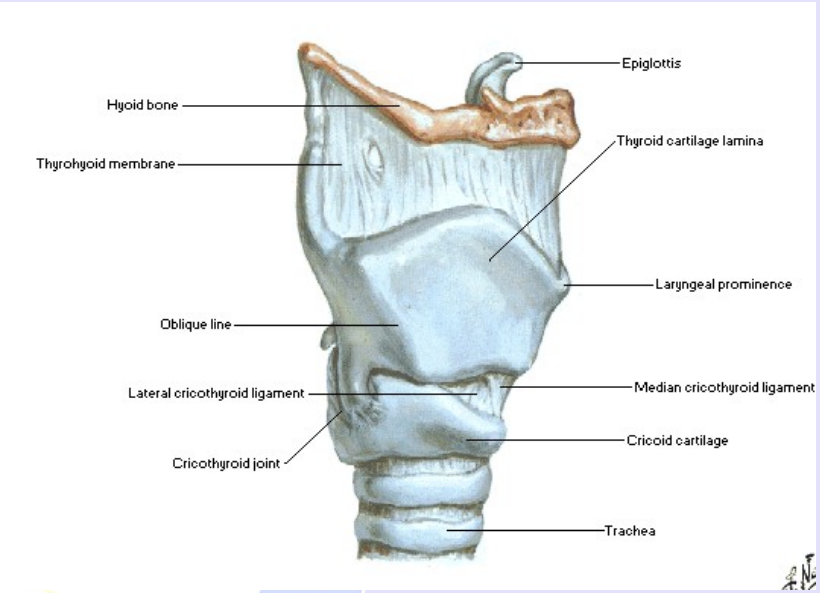
- Rod shaped pieces placed in aryepiglottic folds



JOINTS OF LARYNX

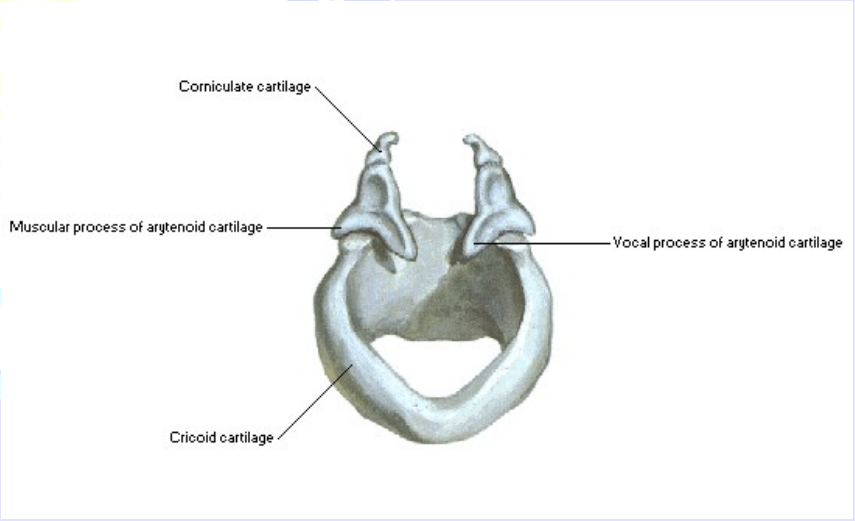
Cricothyroid:

- Synovial
- Between inferior cornu of thyroid cartilage and side of cricoid cartilage
- Rotatory and gliding movements in transverse axis.



Cricoarytenoid:

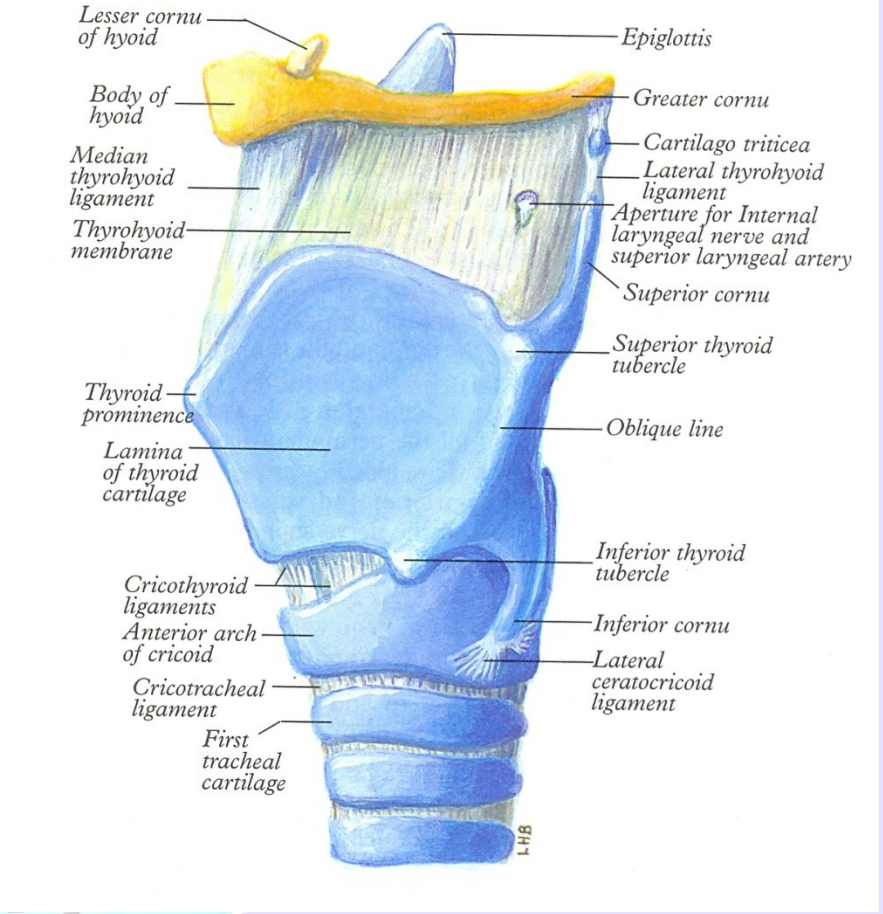
- Synovial
- Between base of arytenoid cartilage and the upper border of lamina of the cricoid cartilage
- Rotatory and gliding movements in vertical axis



MEMBRANES AND LIGAMENTS OF LARYNX

Thyrohyoid membrane:

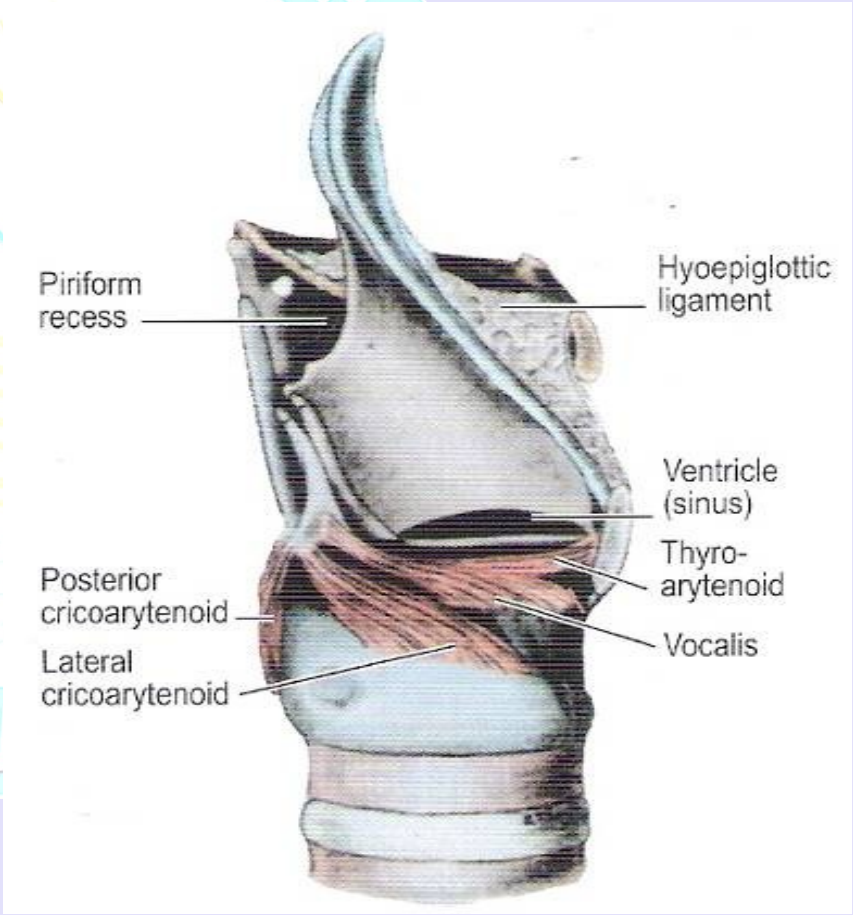
- between thyroid cartilage and hyoid bone
- median and lateral part form **median and lateral thyrohyoid ligament**.
- pierced by internal laryngeal nerve and superior laryngeal vessels.



MEMBRANES AND LIGAMENTS OF LARYNX

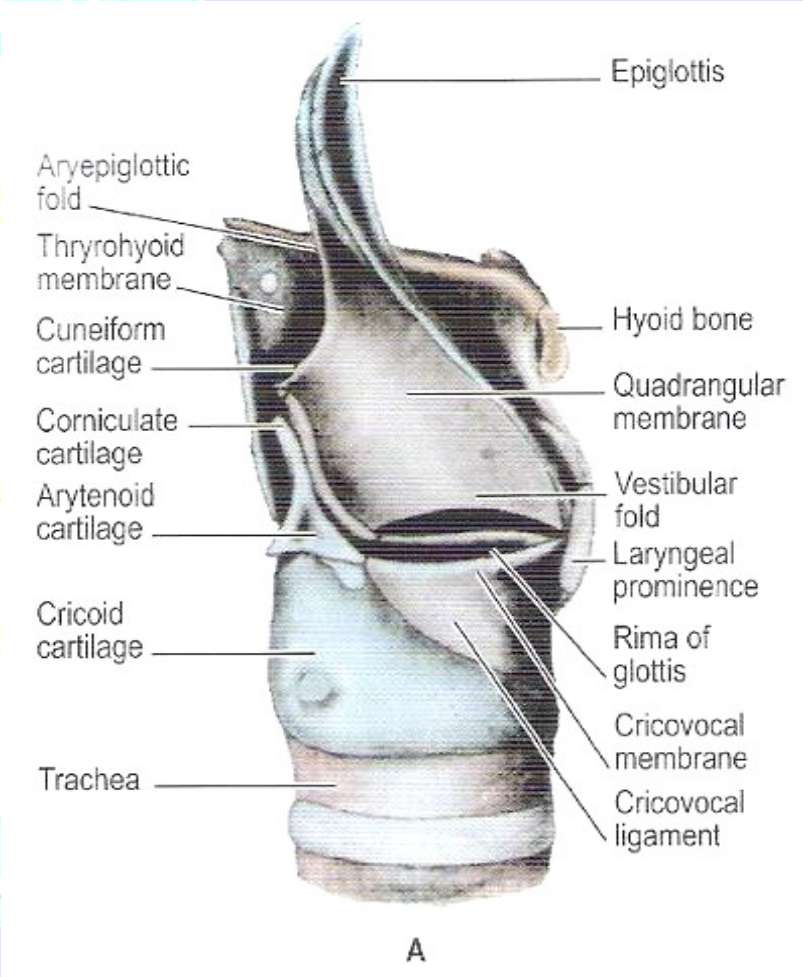
- **Hyoepiglottic lig.:** Between epiglottic cartilage and hyoid bone.

- **Cricotracheal lig.:** Cricoid cartilage and trachea(1ST RING).



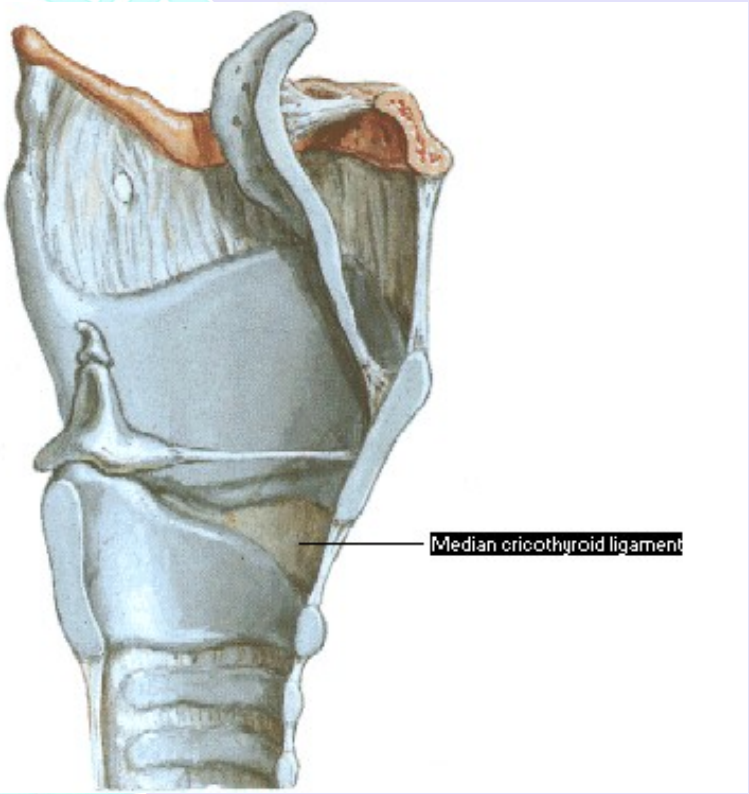
MEMBRANES AND LIGAMENTS OF LARYNX

- QUADRANGULAR MEMBRANE
- Extends between epiglottis and arytenoid cartilages
- Thickened inferior margin forms vestibular ligament



MEMBRANES AND LIGAMENTS OF LARYNX

- Cricothyroid ligament
- Loer border at cricoid upper border
- Upper margin , elastic, forms vocal ligament
- VOCAL LIGAMENT
- Anterior end attached to thyroid cartilage
- Posterior end attached to vocal process arytenoid cartilage



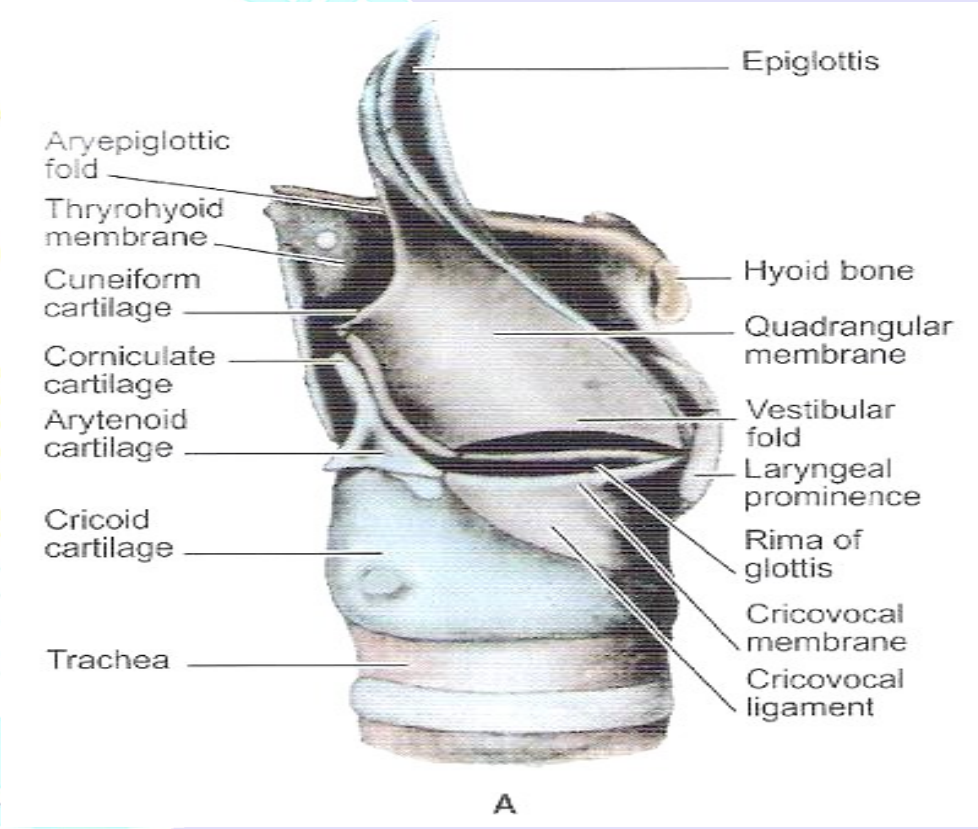
LARYNGEAL FOLD

VESTIBULAR FOLD

- formed by vestibular ligament
- Covered by respiratory epithelium

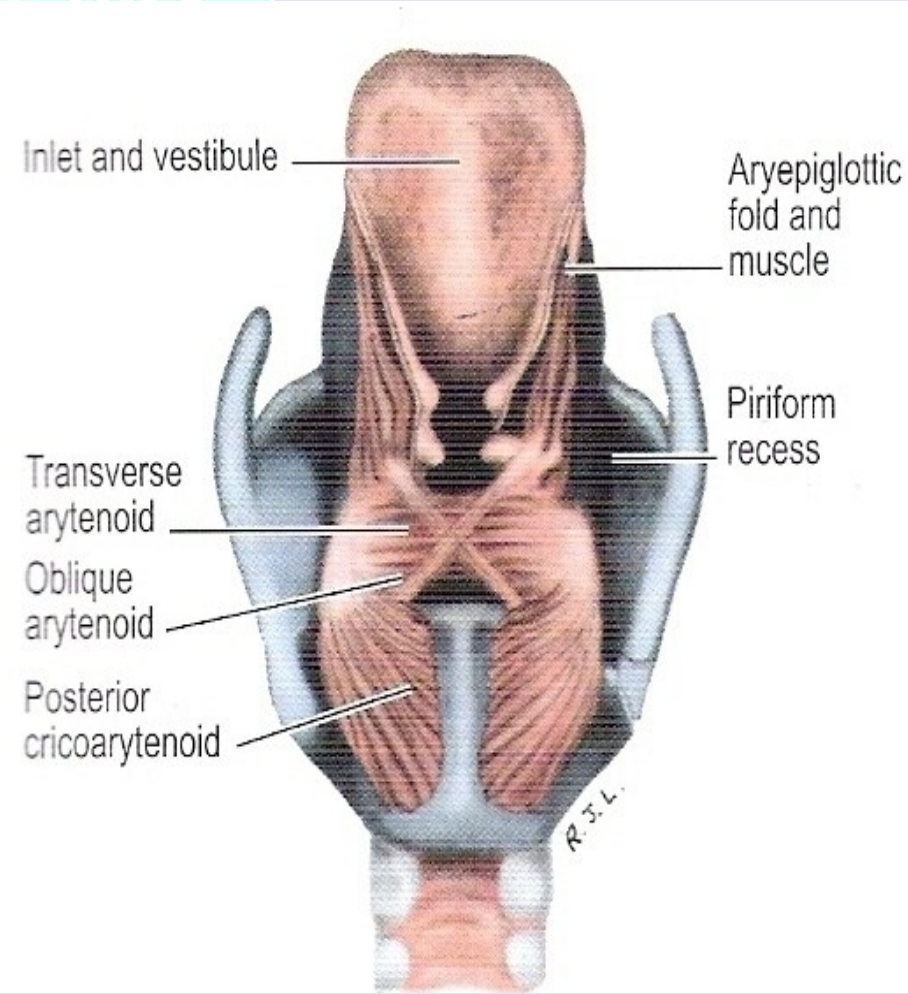
VOCAL FOLD

- Mobile
- Formed by vocal ligament
- Covered by stratified squamous epithelium
- Gap between vocal fold– rima glottis



INLET OF LARYNX

- Backwards and upwards
- Wider in front than behind
- Boundaries
- Anteriorly epiglottis
- Laterally aryepiglottic folds
- Posteriorly arytenoid cartilages

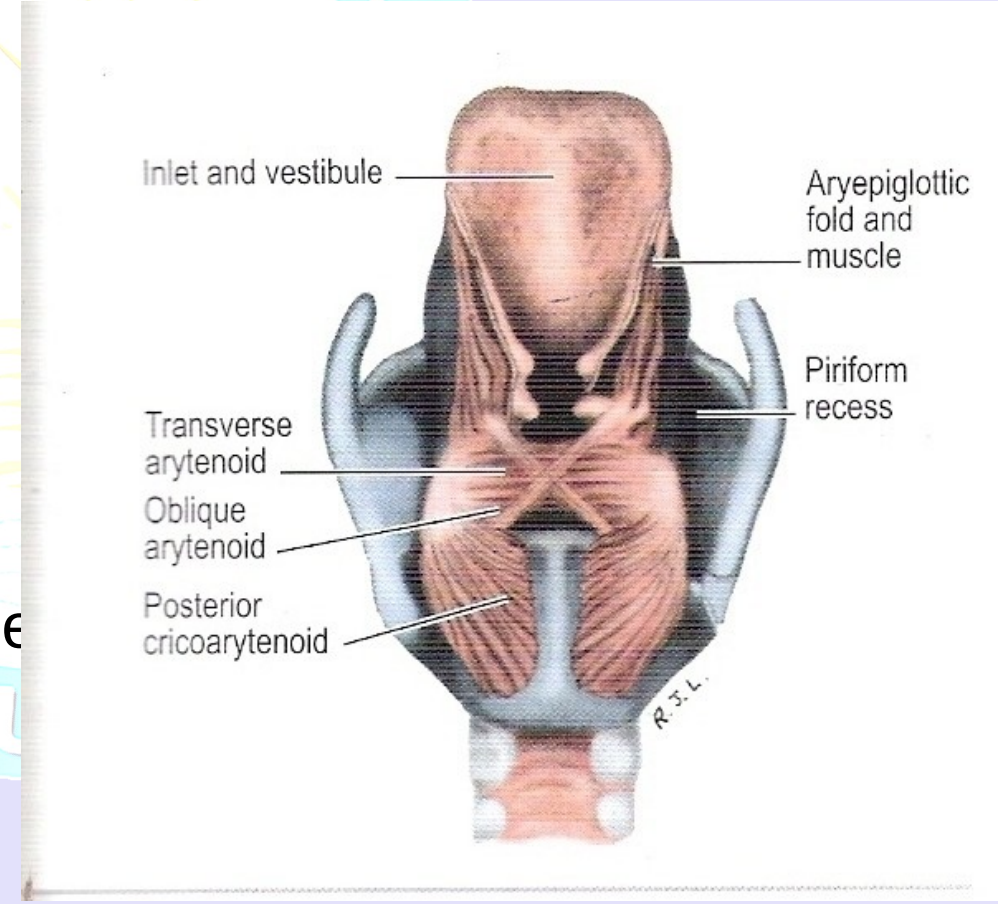


PIRIFORM FOSSA

- Recess on either side of inlet

Boundaries

- Medially
  - Aryepiglottic fold
- Laterally
  - Thyroid cartilage
  - Thyrohyoid membrane

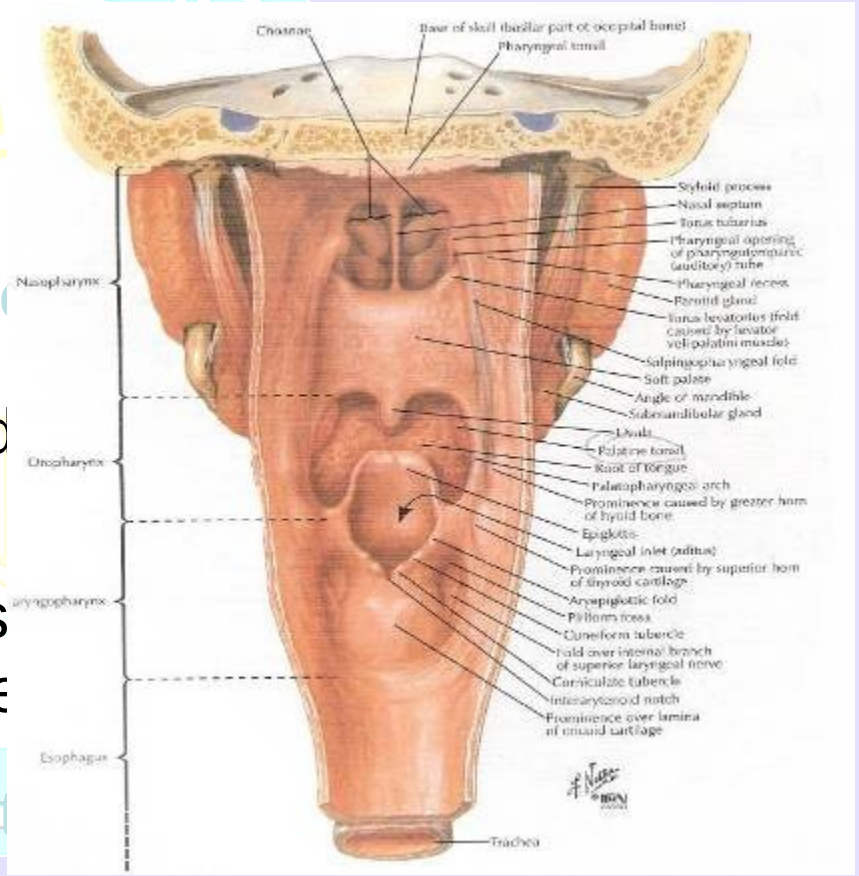


CAVITY OF LARYNX

- Tubular in shape ;lined by mucosa
- Extends from inlet to lower border of cricoid

Boundaries

- Anterior border -mucosa covering the superior margin the epiglottis.
- Lateral borders - mucosal fold (aryepiglottic folds), which enclose the superior margins the quadrangular membranes
- Posterior border in the midline mucosal fold that forms a depression (interarytenoid not between the two corniculate tubercles.

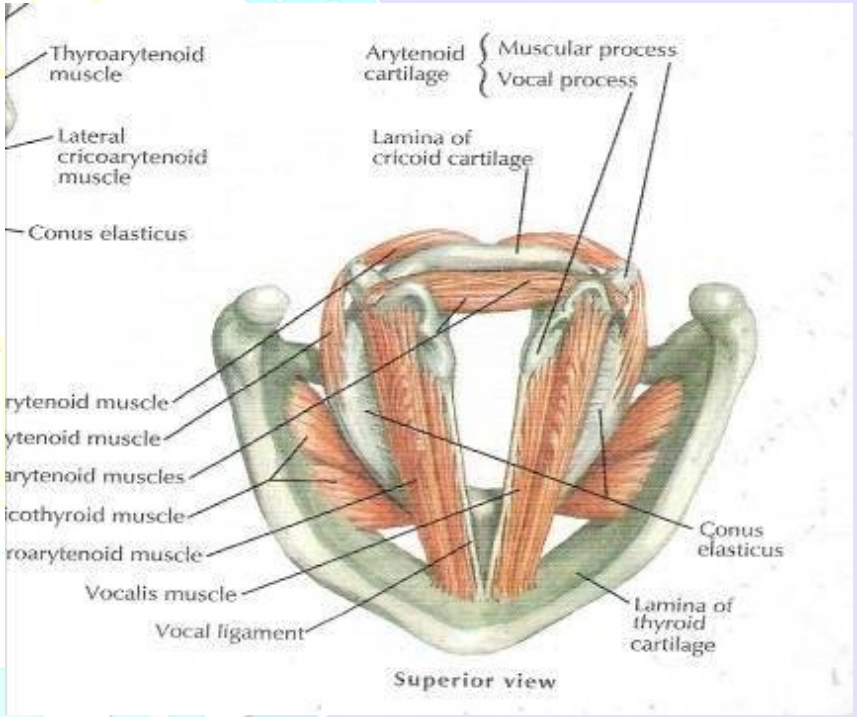


GLOTTIS

- Also known as ***rima glottidis***
- Space between two vocal folds

Boundaries

- Anteriorly
  - Vocal folds
- Posteriorly
  - Medial surface of Arytenoid cartilage
- Narrowest part of larynx
- 2.5cm from front to backin males, less in females

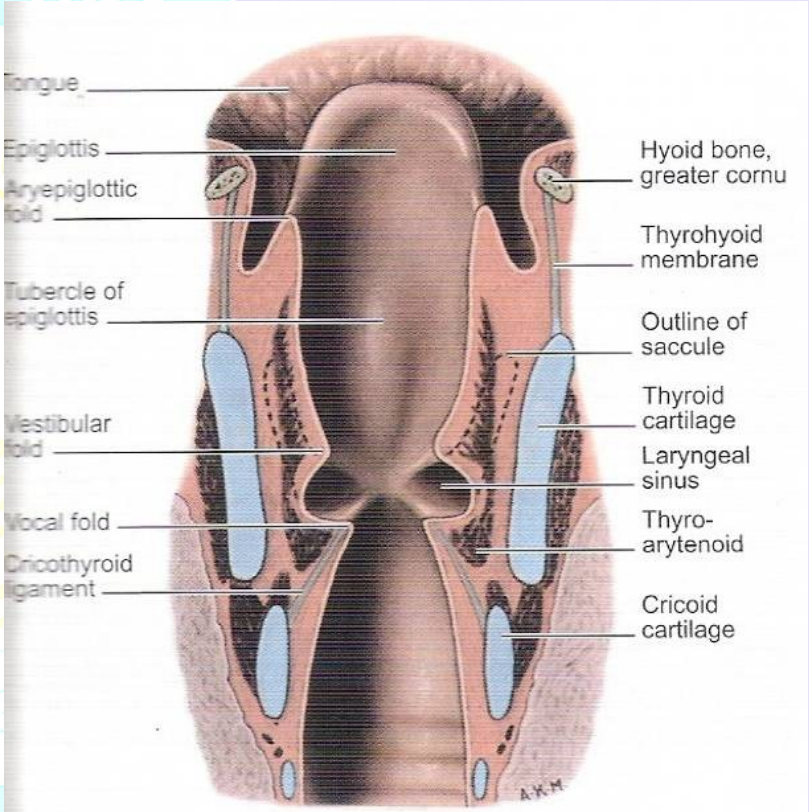


DIVISION OF LARYNGEAL CAVITY

i. The **vestibule** between the laryngeal inlet and the vestibular folds,.

ii. **The middle part** of the laryngeal cavity is very thin and is between the vestibular folds above and the vocal folds below.

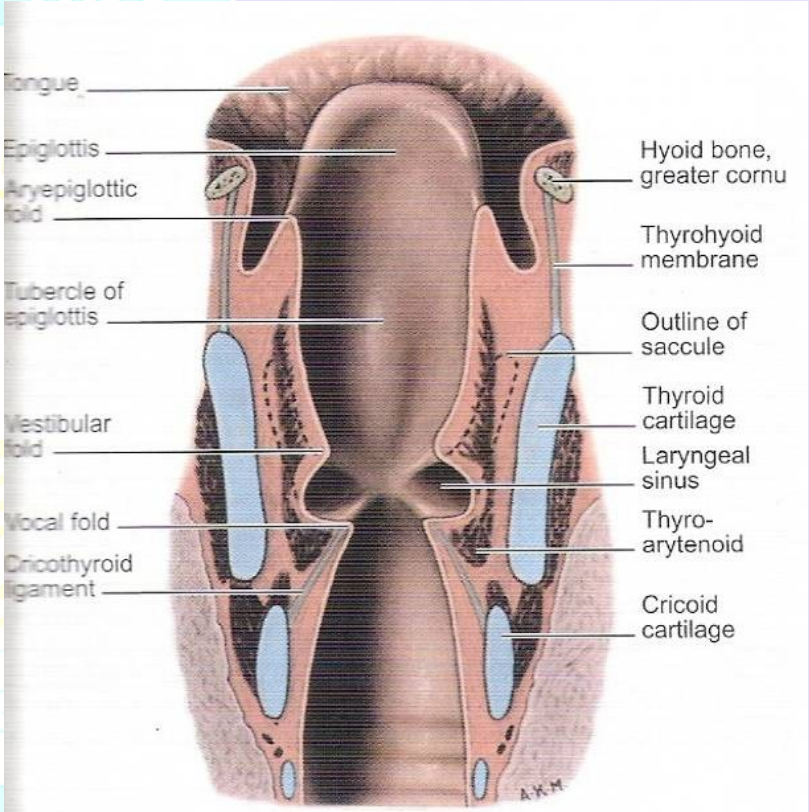
iii. The **infraglottic space** between the vocal folds (and the inferior opening of the larynx.



6.48 Anterior half of the larynx and upper trachea ich have been sectioned in the coronal plane.

SINUS OF LARYNX

- Part between vocal and vestibular fold is **ventricle of larynx** (sinus).
- The sinus of larynx anterior is prolonged as diverticulum **- saccule of larynx** which contains mucous glands for lubrication of vocal folds.



g. 6.48 Anterior half of the larynx and upper trachea ich have been sectioned in the coronal plane.

EXTRINSIC MUSCLES

- Attached to hyoid
- Move larynx by thyrohyoid membrane

Elevators of larynx

- Digastric
- Stylohyoid
- Mylohyoid
- Geniohyoid
- Stylopharyngeus
- Salpingopharyngeus
- Palatopharyngeus

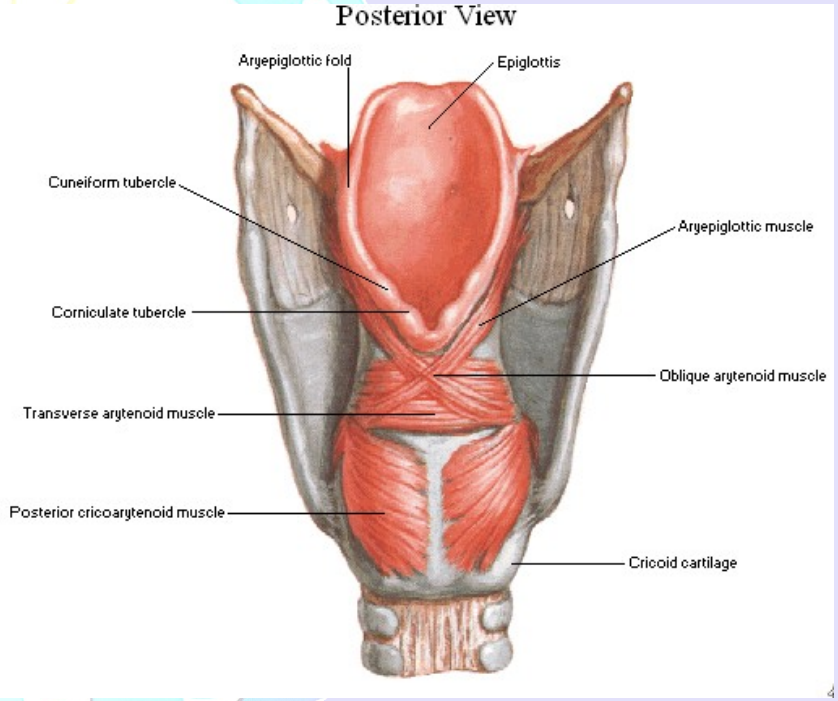
EXTRINSIC MUSCLES

Depressor of larynx

- Sterno thyroid
- Sternohyoid
- Omohyoid

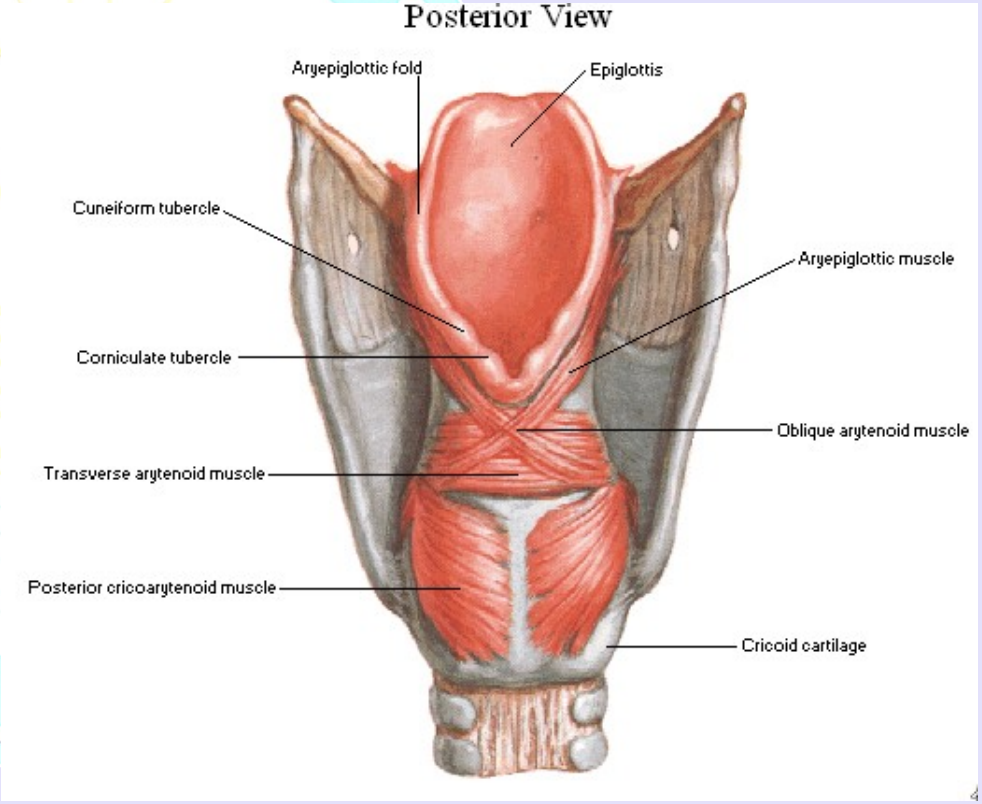
INTRINSIC MUSCLES

- Cricothyroid
- **Lat**. And post. Cricoarytenoid,
- Transverse arytenoid
- Oblique arytenoid
- Thyro arytenoid (vocalis)
- Thyro epiglottic



FUNCTIONS OF INTRINSIC MUSCLES

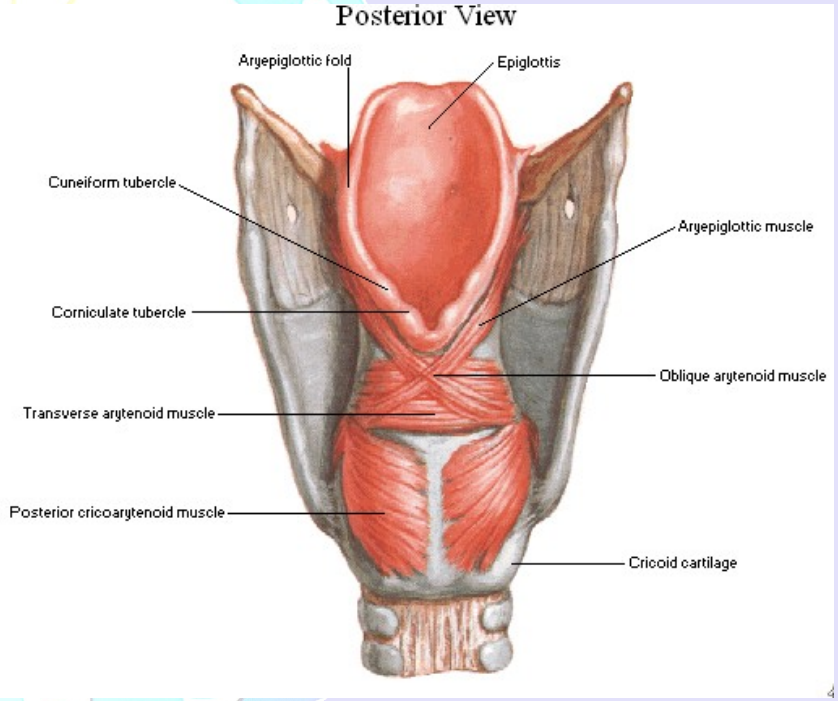
- **MUSCLES CONTROLLING INLET**
- Narrowing-oblique arytenoid
- Widening thyroepiglottic



FUNCTIONS OF INTRINSIC MUSCLES

- MUSCLES MOVING VOCAL CORDS

- Tensor- cricothyroid
- Relaxor- thyroarytenoid
- Adductor-lateral cricoarytenoid
- Abductor- posterior cricoarytenoid
- Approximate arytenoid- transverse arytenoid



MUCOSA OF LARYNX

stratified squamous noncornified epithelium

- laryngeal inlet
- most of the epiglottis
- true vocal cords.

Typical respiratory epithelium

- the rest of the larynx.

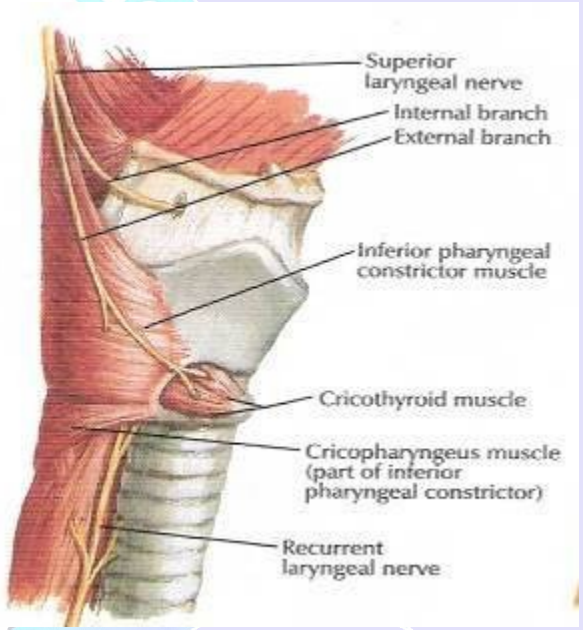
NERVE SUPPLY OF LARYNX

MOTOR

- Recurrent laryngeal nerve except for cricothyroid (external laryngeal nerve)

SENSORY

- Above vocal cords
- Internal laryngeal branch of sup. Laryngeal of vagus
- Below vocal cords
- Recurrent laryngeal



BLOOD SUPPLY OF LARYNX

Upper half

- Superior laryngeal branch of superior throid artery

Lower half

- Inferior laryngeal branch of inferior thyroid artery
- Lymphatics- drain into deep cervical group



Thanks